n-Valued Groups, Kronecker Sums, and Wendt Matrices*

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A Brief History

In 1971, V. M. Buchstaber and S. P. Novikov proposed a construction *motivated by the theory of characteristic classes* [BN71].

This construction describes a multiplication in which *the product of any pair of elements is a multiset of n points* [Buc06].

An axiomatic definition of n-valued groups and the first results of their algebraic theory were obtained in a subsequent series of works by V. M. Buchstaber.

Currently, the theory of n-valued (formal, finite, discrete, topological, and algebro-geometric) groups and their applications in various areas of mathematics and mathematical physics are being developed by a number of authors.

n-Valued Monoids

An *n-valued monoid* is a set X equipped with an operation

$$* : X \times X \to \mathrm{Sym}^n(X)$$

where $\mathrm{Sym}^n(X) = X^{\times n}/S_n$ is the set of unordered n-tuples of elements of X:

- *Associativity.* The n^2-multisets

$$[x * w \mid w \in y * z],$$
$$[w * z \mid w \in x * y]$$

  coincide.
- *Unit.* There exists an element $e \in X$ such that

$$e * x = x * e = [x, x, \ldots, x]$$

  for every $x \in X$.

n-Valued Groups

An *n-valued group* X is an n-valued monoid equipped with an inverse map

$$\mathrm{inv} : X \to X$$

that is, a map such that for every $x \in X$

$$x * \mathrm{inv}(x) \ni e, \quad \mathrm{inv}(x) * x \ni e.$$

The notions of homomorphisms, commutativity, kernels, and related concepts admit natural generalizations from the context of 1-valued groups to that of n-valued groups.

Groups $\mathbb{X}_n$

For each $n \geqslant 3$, there exists a commutative n-valued group $\mathbb{X}_n$ on the set $\{0, 1, 2\}$ with unit 0 [Kor25, Example 3]:

$$1 * 1 = [0, \underbrace{2, \ldots, 2}_{n-1}]$$

$$1 * 2 = [0, 1, \underbrace{2, \ldots, 2}_{n-2}]$$

$$2 * 2 = [0, 1, 1, \underbrace{2, \ldots, 2}_{n-3}]$$

The inverse map inv may be defined in four different ways: $\mathrm{Inv}(1) = \mathrm{Inv}(2) = \{1, 2\}$.

Coset (Étale) Construction

Given 1-valued group G with multiplication μ_0, identity e_G, and inverse $\mathrm{inv}_G(u) = u^{-1}$, and given $H < \mathrm{Aut}(G)$, $|H| = n$.

Consider *the set of orbits* $X := G/H$ with projection $\pi : G \to X$ and an n-valued multiplication

$$\mu : X \times X \to \mathrm{Sym}^n(X)$$

$$\mu(x, y) = [\pi(\mu_0(u, h(v))) \mid h \in H]$$

where $u \in \pi^{-1}(x)$ and $v \in \pi^{-1}(y)$.

Theorem (Buchstaber, [Buc06])

The multiplication μ defines an n-valued multiplication on X with identity $e_x = \pi(e_G)$ and inverse $\mathrm{inv}_X(x) = \pi(\mathrm{inv}_G(u))$, where $u \in \pi^{-1}(x)$.

Groups $\mathbb{G}_n$

Consider the operation $* : \mathbb{C} \times \mathbb{C} \to \mathrm{Sym}^n(\mathbb{C})$ that assigns to each pair of complex numbers x and y the multiset

$$x * y = [(\sqrt[n]{x} + \varepsilon^r \sqrt[n]{y})^n \mid r = 1 \ldots n]$$

of n complex numbers, where ε is a primitive nth root of unity, and $\sqrt[n]{\cdot}$ denotes a fixed root branch.

The operation $*$, the identity element 0, and the inverse $\mathrm{inv}(x) = (-1)^n x$ endow the set of complex numbers $\mathbb{C}$ with the structure of a *commutative n-valued group* $\mathbb{G}_n$.

Groups $\mathbb{G}_n$

$\mathbb{G}_n$'s are examples of coset groups.

Let G be the additive subgroup of $\mathbb{C}$. For a fixed a primitive nth root of unity ε, consider an automorphism

$$\varphi : z \mapsto \varepsilon z$$

We have the projection

$$\pi : \mathbb{C} \to \mathbb{C}\,/\langle\varphi\rangle \cong \mathbb{C}$$
$$z \mapsto z^n$$

So, in this case we get

$$x * y = [(\sqrt[n]{x} + \varepsilon^r \sqrt[n]{y})^n \mid r = 1 \ldots n]$$

Groups $\mathbb{G}_n$

§ 2b. Обратим теперь внимание на то обстоятельство, что степенная система $B^{-1}(k^2B(x)) = k^2\Psi^k(u\bar{u})$ связана со своеобразной «двузначной формальной группой»

$$F^{\pm}(x, y) = B^{-1}((\sqrt{B(x)} \pm \sqrt{B(y)})^2),$$

в которой операция возведения в степень однозначна, а именно, $B^{-1}(k^2B(x)) = F^{\pm}(\underbrace{x, \ldots, x}_{k \text{ штук}})$. Если $x = u\bar{u}$, $y = v\bar{v}$, то $F^{\pm}(x, y) = \{|f(u, v)|^2; f(u,\bar{v})|^2\}$ и для $U(1)$-расслоений ξ, η над $CF^\infty \times CP^\infty$, где $u = \sigma_1(\xi)$, $v = \sigma_1(\eta)$, мы имеем $F^{\pm}(x, y) = \{\sigma_2(\xi\eta + \bar{\xi}\bar{\eta}); \sigma_2(\xi\bar{\eta} + \bar{\xi}\eta)\}$, $x = \sigma_2(\xi + \bar{\xi})$, $y = \sigma_2(\eta + \bar{\eta})$.

Excerpt from [BN71]

Non-coset Groups

For every n-valued group on a set X, in addition to the inverse map $\mathrm{inv} : X \to X$, we define a multivalued inverse map $\mathrm{Inv} : X \to \mathscr{P}(X)$ taking values in the set of all finite subsets of X:

$$\mathrm{Inv}(x) = \{y \in X \mid x * y \ni e, \quad y * x \ni e\}.$$

An n-valued group has a uniquely defined inverse map if and only if $\mathrm{inv} = \mathrm{Inv}$. We call such groups *reversible*; otherwise, we call them *non-reversible*.

Every coset group is reversible. So, $\mathbb{X}_n$ is not a coset group.

Combinatorial Algebras

An n-valued group $(X, *, \mathrm{inv})$ is called ★*-involutive* if the inverse map $x \mapsto \mathrm{inv}(x)$ is an involution satisfying the following conditions:

1. For any $x \in X$, one has $x * y \ni e$ if and only if $y = \mathrm{inv}(x)$.
2. $m(x) = m(\mathrm{inv}(x))$, where $m(x) = m^e_{x,\mathrm{inv}(x)}$ is the multiplicity of e in the product $x * \mathrm{inv}(x)$.
3. $\mathrm{inv}(x * y) = \mathrm{inv}(y) * \mathrm{inv}(x)$ for all $x, y \in X$.

The category of combinatorial algebras and the category of multivalued involutive groups are equivalent [BVEP96].

The Classification of n-Valued Monoids

The complete classification of all $\bigstar$-involutive n-valued groups of order 3 was achieved in [Pon24, GPV25], based on the classification of all single-valued finite groups of rank 3.

A full classification of n-valued monoids and groups of order 3 has been recently obtained by Kornev in *arXiv:2508.04454* [Kor25].

As a corollary, *there are non-$\bigstar$-involutive non-coset n-valued groups*.

Back to the Group $\mathbb{G}_n$

Recall the operation $* : \mathbb{C} \times \mathbb{C} \to \mathrm{Sym}^n(\mathbb{C})$

$$x * y = [(\sqrt[n]{x} + \varepsilon^r \sqrt[n]{y})^n \mid r = 1 \dots n]$$

where ε is a primitive nth root of unity, and $\sqrt[n]{\cdot}$ denotes a fixed root branch.

The operation $*$, the identity element 0, and the inverse $\mathrm{inv}(x) = (-1)^n x$ endow the set of complex numbers $\mathbb{C}$ with the structure of a *commutative n-valued group* $\mathbb{G}_n$.

Introducing Polynomials $p_n(z; x, y)$

Introduce the following symmetric polynomials:

$$p_n(z; x, y) = \prod_{w \in \mathrm{inv}(x) * \mathrm{inv}(y)} (z - w)$$

$$\begin{aligned}
p_1 &= \sigma_1, \\
p_2 &= \sigma_1^2 - 4\sigma_2, \\
p_3 &= \sigma_1^3 - 27\sigma_3, \\
p_4 &= \sigma_1^4 - 2^3\sigma_1^2\sigma_2 + 2^4\sigma_2^2 - 2^7\sigma_1\sigma_3, \\
p_5 &= \sigma_1^5 - 5^4\sigma_1^2\sigma_3 - 5^5\sigma_2\sigma_3, \\
p_6 &= \sigma_1^6 - 2^2 \cdot 3\sigma_1^4\sigma_2 - 2 \cdot 3^4 \cdot 17\sigma_1^3\sigma_3 + 2^4 \cdot 3\sigma_1^2\sigma_2^2 \\
&\quad - 2^3 \cdot 3^4 \cdot 19\sigma_1\sigma_2\sigma_3 - 2^6\sigma_2^3 + 3^3 \cdot 19^3\sigma_3^2
\end{aligned}$$

where $\sigma_j's$ are elementary symmetric polynomials in x, y, z.

Introducing Polynomials $p_n(z; x, y)$

For $p_{18}(z; x, y)$ we have:

$$
\begin{aligned}
&(18, 0, 0) \to 1\\
&(16, 1, 0) \to -2^2 3^2\\
&(15, 0, 1) \to -2^1 3^5 43^1 293^1 13339^1\\
&(14, 2, 0) \to 2^6 3^2\\
&(13, 1, 1) \to -2^1 3^5 39079^1 30478663^1\\
&(12, 3, 0) \to -2^8 3^1 7^1\\
&(12, 0, 2) \to 3^4 167^1 58369^1 1702940402507^1\\
&(11, 2, 1) \to -2^2 3^6 57769225879741^1\\
&(10, 4, 0) \to 2^9 3^2 7^1\\
&(10, 1, 2) \to -2^2 3^5 7^1 97^1 36913^1 180317^1 3375001057^1\\
&(9, 3, 1) \to -2^1 3^5 23^1 144589^1 5245247209^1\\
&(9, 0, 3) \to -2^1 3^8 5^1 12713^1 76919^1 37764598382689403^1\\
&(8, 5, 0) \to -2^{11} 3^2 7^1\\
&(8, 2, 2) \to 3^5 67589^1 6265409413034952103 51^1\\
&\textit{etc.}
\end{aligned}
$$

where, e. g., $(16, 1, 0) \to -2^2 3^2$ means that the coefficient of $x^{16} y$ equals $-2^2 3^2$.

Introducing Polynomials $p_n(z; x, y)$

One can define the n-valued group $\mathbb{G}_n$ via the roots of polynomials:

$$\widetilde{p}_n = p_n(z; (-1)^n x, (-1)^n y)$$

$$x * y = [z \mid z \text{ is a root of } \widetilde{p}_n(z; x, y)]$$

These polynomials were first introduced in [BN71] in the context of the theory of n-valued groups.

The problem of describing the structure of the coefficients of the polynomials p_n was posed in [Buc06], where also the explicit form of $p_n(z; x, y)$ was given for $1 \leqslant n \leqslant 7$.

Kronecker Sums

Recall the following definition:

Let A and B be arbitrary square matrices over the field $\mathbb{k}$ of orders m and n, respectively.

The *Kronecker sum* $A \boxplus B$ of the matrices $A \in \mathrm{Mat}_{m\times m}(\mathbb{k})$ and $B \in \mathrm{Mat}_{n\times n}(\mathbb{k})$ is the $mn \times mn$ matrix

$$A \boxplus B = A \otimes I_n + I_m \otimes B.$$

This notion has many applications in various fundamental and applied fields of science, including differential equations, quantum computing, machine learning, robotics, and others.

Kronecker Sums

Proposition (Stephanos, [Sté00])

Let $A \in \mathbb{C}^{m\times m}$, $B \in \mathbb{C}^{n\times n}$, *and* $p(x,y) = \sum_{j,k=0}^{\ell} c_{jk} x^j y^k \in \mathbb{C}[x,y]$.

Consider the matrix

$$p(A,B) = \sum_{j,k=0}^{\ell} c_{jk} A^j \otimes B^k$$

Let $\lambda_1, \ldots, \lambda_m$ *be the eigenvalues of the matrix* A, $\mu_1, \ldots, \mu_n$ *the eigenvalues of* B.

Then the eigenvalues of the matrix $p(A,B)$ *are the* mn *numbers* $p(\lambda_r, \mu_s)$, *where* $r = 1, \ldots, m$ *and* $s = 1, \ldots, n$.

Frobenius Companion Matrices

Let $f(t)$ be $t^m + a_{m-1}t^{m-1} + \cdots + a_0 \in \mathbb{C}[t]$.

The *Frobenius companion matrix* $F(a_{m-1}, \ldots, a_0)$ of the polynomial $f(t)$ is

$$\begin{pmatrix} 0 & 0 & 0 & 0 & -a_0 \\ 1 & 0 & 0 & 0 & -a_1 \\ 0 & 1 & 0 & 0 & -a_2 \\ 0 & 0 & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 1 & -a_{m-1} \end{pmatrix}.$$

$p_n(z; x, y)$ and Kronecker Sums

Proposition (Buchstaber, Kornev, 2025)

The following statements hold for each natural n:

1. $p_n(z; x, y)$ *is symmetric in the variables* x, y, *and* z.

2. $$p_n(z^n; x, y) = \chi(F(\underbrace{0, \ldots, 0, (-1)^{n+1}x}_{n}) \boxplus F(\underbrace{0, \ldots, 0, (-1)^{n+1}y}_{n}); z)$$

3. *The polynomial* $p_n(z; x, y)$ *is a homogeneous polynomial in the variables* x, y, *and* z *of degree* n *with integer coefficients.*

$p_n(z; x_1, \ldots, x_m)$ and Kronecker Sums

Theorem (Buchstaber, Kornev, 2025)

The following statements hold for each natural n:

1. $p_n(z; x_1, \ldots, x_m)$ *is symmetric in all variables.*
2. *For all complex z, $x_1, \ldots, x_m$, we have the equality*

$$p_n(z^n; x_1, \ldots, x_m) =$$

$$\chi(F(\underbrace{0, \ldots, 0, (-1)^{n+1} x_1}_{n}) \boxplus \cdots \boxplus F(\underbrace{0, \ldots, 0, (-1)^{n+1} x_m}_{n}); z).$$

3. $p_n(z; x_1, \ldots, x_m)$ *is homogeneous in the variables z, x_1, …, x_m of degree n^{m-1} with integer coefficients.*

θ-Circulants

For any elements $a_0, \ldots, a_{n-1} \in \Bbbk$, introduce a *$\theta$-circulant matrix*:

$$\mathrm{Circ}_\theta(a_0, \ldots, a_{n-1}) = \begin{pmatrix} a_0 & \theta a_1 & \theta a_2 & \cdots & \theta a_{n-1} \\ a_{n-1} & a_0 & \theta a_1 & \ddots & \vdots \\ a_{n-2} & a_{n-1} & a_0 & \ddots & \theta a_2 \\ \vdots & \ddots & \ddots & \ddots & \theta a_1 \\ a_1 & \cdots & a_{n-2} & a_{n-1} & a_0 \end{pmatrix}$$

Wendt Matrices

This class contains *Wendt matrices* W_n of the form

$$\begin{pmatrix} 1 & \binom{n}{1} & \binom{n}{2} & \cdots & \binom{n}{n-1} \\ \binom{n}{n-1} & 1 & \binom{n}{1} & \cdots & \binom{n}{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \binom{n}{1} & \binom{n}{2} & \binom{n}{3} & \cdots & 1 \end{pmatrix}$$

Proposition (Wendt, [Wen94])

Let p be an odd prime and $q = 2kp + 1$ be prime for some $k \geqslant 1$. Then there exist integers x, y, z, not divisible by q, such that

$$x^p + y^p + z^p \equiv 0 \pmod{q}$$

if and only if q divides $\det W_{2k}$.

Wendt Matrices

Following tradition, we say that the *first case of Fermat's theorem is true for the odd prime exponent p* when:

if x, y, z are (non-zero) integers, not multiples of p, then $x^p + y^p \neq z^p$.

Wendt Matrices

Theorem (Germain, Legendre, 1823, [Rib99])

Let p and q be distinct odd prime numbers, and suppose the following conditions hold:

- *If a, b, c are integers and $a^p + b^p + c^p \equiv 0 \pmod{q}$, then $q \mid abc$.*
- *There is no integer m such that $m^p \equiv p \pmod{q}$.*

Then the first case of Fermat's Last Theorem holds for the exponent p.

$p_n(z; x, y)$ and Wendt Matrices

Theorem (Buchstaber, Kornev, 2025)

The polynomial $p_n(z; x, y)$ defining the n-valued multiplication is the determinant of a y-circulant $n \times n$ matrix:

$$\mathrm{Circ}_y \left(w^n + (-1)^{n+1} x + y, \binom{n}{1} w, \ldots, \binom{n}{n-1} w^{n-1} \right)$$

where $w^n = z$.

These matrices generalize the Wendt matrices (the substitution $x = (-1)^n$, $y = z = 1$).

$p_n(z; x, y)$ and Wendt Matrices

$$p_2 = \begin{vmatrix} w^2 - x + y & 2wy \\ 2w & w^2 - x + y \end{vmatrix}$$

$$p_3 = \begin{vmatrix} w^3 + x + y & 3yw & 3yw^2 \\ 3w^2 & w^3 + x + y & 3yw \\ 3w & 3w^2 & w^3 + x + y \end{vmatrix}$$

$$p_4 = \begin{vmatrix} w^4 - x + y & 4yw & 6yw^2 & 4yw^3 \\ 4w^3 & w^4 - x + y & 4yw & 6yw^2 \\ 6w^2 & 4w^3 & w^4 - x + y & 4yw \\ 4w & 6w^2 & 4w^3 & w^4 - x + y \end{vmatrix}$$

The Coefficients of $p_n(z; x, y)$

Theorem (Buchstaber, Kornev, 2025)

For prime $n \geqslant 5$, the polynomial

$$p_n(z; x, y) - (x + y + z)^n$$

is divisible by n^4xyz.

This result follows from the above observations and from the following:

Theorem (Wolstenholme, [Wol62])

$$\binom{2n-1}{n-1} - 1$$

is divisible by n^3 for primes $n \geqslant 5$.

The Irreducibility of $p_n(z; x, y)$

Theorem (Buchstaber, Kornev, 2025)

The polynomials $p_n(z; x_1 \dots x_m)$ in the variables $z_1 \dots x_1 \dots x_m$ are irreducible over $\mathbb{C}$ for each positive integer n.

Theorem (Buchstaber, Kornev, 2025)

Fix a positive integer $n \geqslant 2$. Let the positive integer pairwise coprime numbers $x_1 \dots x_m$ be such that $(\sqrt[n]{x_j})^k$ is irrational for all positive integer $k < n$ for each $j = 1 \dots m$. Then the polynomial $p(z; x_1 \dots x_m)$ in the variable z is irreducible over $\mathbb{Q}$.

p_n and Discriminants

Theorem ([GRS24], direct proof in [BK25])

The discriminant $\operatorname{disc}(P_{x,y,z})$ *of the polynomial*

$$P_{x,y,z}(T) = ((-1)^n x T^{n-1} + (-1)^n y)(1+T)^{n-1} - T^{n-1} z$$

with respect to the variable T*, which is a polynomial of degree* $4n-6$*, is related to the polynomial* $p_n(z; x, y)$ *as follows:*

$$(-1)^n (n-1)^{2n-2} (xyz)^{n-2} p_n(z; x, y) = \operatorname{disc}_T(P_{x,y,z}(T))$$

for each $n \geqslant 2$*.*

$\operatorname{disc}(P_{x,y,z})$ is the discriminant of the homogenized polynomial

$$(u+v)^{n-1}(xu^{n-1} + yv^{n-1}) - (uv)^{n-1} z$$

with respect to the variables u and v.

p_n and Discriminants

We get the following:

Corollary (Helou, [Hel97])

For each odd $n \geqslant 3$, the discriminant $\operatorname{disc} Q(t)$ of the polynomial $Q(t) = (1+t)^n - t^n - 1$ and the determinant $\det W_{n-1}$ of the Wendt matrix are related by the following equality:

$$\operatorname{disc} Q(t) = (-1)^{\frac{n-1}{2}} n^{n-2} \det W_{n-1}.$$

p_n and Discriminants

Consider the homogeneous polynomial

$$P_x(u) = (u_1 \dots u_m)^{n-1} \left(z + (-1)^n \left(\sum_{j=1}^{m} u_j \right)^{n-1} \cdot \sum_{j=1}^{m} \frac{x_j}{u_j^{n-1}} \right)$$

of degree $m(n-1)$ in the variables $u_1 \dots u_m$.

The discriminant $\mathrm{disc}(P_x(u))$ of the polynomial $P_x(u)$ is a homogeneous polynomial of degree $m(m(n-1)-1)^{m-1}$.

p_n and Discriminants

From [GRS24, Theorem 3.2], one gets

Theorem ([GRS24])

The discriminant $\operatorname{disc} P_x(u)$ *of the homogeneous polynomial* $P_x(u)$*, which has degree* $m(n-1)$ *in the variables* $u_1 \dots u_m$*, is divisible by certain powers of the monomial* $zx_1 \dots x_m$ *and the polynomial* $p_n(z; x_1 \dots x_m)$*.*

n-Valued Groups and Representations

Let a_{jk}^{ℓ} be the multiplicities in the decomposition of the tensor product $\rho_j \otimes \rho_k$ into a sum of irreducible finite dimensional complex representations $\{\rho_j\}$ of a finite group G:

$$\rho_j \otimes \rho_k = \sum_{\ell=0}^{k} a_{jk}^{\ell} \rho_\ell .$$

Theorem (Buchstaber, [Buc06])

Let $d_j = \dim \rho_j$, $n = lcm(d_j d_k \mid j, k = 0 \ldots k)$, *and* $m_{jk}^{\ell} = n a_{jk}^{\ell} d_\ell / (d_j d_k)$. *Then on the set* $\left\{ \frac{\rho_j}{\dim \rho_j} \right\}$ *there is a commutative n-valued group structure* $\mathbb{G}_{\otimes}^{\mathrm{rep}}$, *such that the element* $\rho_\ell / \dim \rho_\ell$ *appears in the n-set* $\rho_j / \dim \rho_j * \rho_k / \dim \rho_k$ *with multiplicity* m_{jk}^{ℓ}.

n-Valued Groups and Representations

For the Kronecker sum, we have:

$$\frac{\rho_j}{d_j} + \frac{\rho_k}{d_k} = \frac{\rho_j \boxplus \rho_k}{d_k d_j}.$$

Theorem (Buchstaber, Kornev, 2025)

Let G be a finite group. The set $\mathbb{G}^{\mathrm{rep}}_{\boxplus}$, equipped with the above operation, neutral element 0, and formal inverse $\mathrm{inv}(\psi/\dim\psi) = -\psi/\dim\psi$, is a 1-group isomorphic to $\mathbb{Z}^N$, where N is the number of conjugacy classes of the group G.

Algebraic n-Valued Groups

Consider a polynomial over $\mathbb{C}$:

$$P(z; x, y) = \sum_{k=0}^{n} F_{n-k}(x, y) z^k$$

where $F_{n-k}(x, y) \in \mathbb{C}[x, y]$ and

$$F_0(0, 0) = 1 \tag{1}$$

Introduce an operation $*$, such that for $x, y \in \mathbb{C}$ with $F_0(x, y) \neq 0$:

$$x * y = [z \mid P(z; x, y) = 0]$$

In this context, let us formulate the axioms from the definition of n-valued groups.

Algebraic n-Valued Groups

From the *neutral element axiom* with $x = 0$, it follows that

$$P(z; 0, y) = F_0(0, y)(z - y)^n \tag{2}$$

The *inverse element axiom*:

$$F_n(x, \mathrm{inv}(x)) = F_n(\mathrm{inv}(x), x) = 0 \tag{3}$$

Denote by $z_j = z_j(x, y)$ the roots of P. Then the *associativity condition*:

$$\prod_{j,k=1}^{n} (t - z_j(z_k(x, y), z)) = \prod_{j,k=1}^{n} (t - z_j(x, z_k(y, z))) \tag{4}$$

Note that the coefficients of the polynomials in the formal variable t on both sides of equation (4) are polynomials in x, y, and z.

Algebraic n-Valued Groups

An *n-valued algebraic group* is an algebraic variety X with an n-valued multiplication defined on a dense subset $Y \subset X \times X$ in the Zariski topology.

An *n-algebraic n-valued group* on $\mathbb{C}$ is the set $\mathbb{C}$ with

$$\begin{aligned} * : (\mathbb{C} \times \mathbb{C}) \backslash \{(x, y) \mid F_0(x, y) = 0\} &\longrightarrow \mathrm{Sym}^n(\mathbb{C}) \\ (x, y) &\longmapsto [\text{roots of } P(z; x, y) = 0] \end{aligned}$$

such that each of the variables x and y appears in it with degree at most n, and the polynomial satisfies conditions (1)–(4).

We denote the corresponding group by $\mathbb{G}(P)$.

n-Algebraic n-Valued Groups: Examples

1. The polynomial $P(z; x, y) = z - x - y$ defines a 1-algebraic 1-valued group on $\mathbb{C}$ with the usual addition as the operation.

2. Consider
$$P(z; x, y) = -x - y + z + \alpha xyz,$$
where $\alpha \in \mathbb{C}$. Then we have a 1-algebraic 1-valued commutative group on $\mathbb{C}$ with the operation
$$x * y = \frac{x + y}{1 + \alpha xy}.$$

$$\begin{pmatrix} 1 & x \\ \alpha x & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & y \\ \alpha y & 1 \end{pmatrix} = (1 + \alpha xy) \begin{pmatrix} 1 & \frac{x+y}{1+\alpha xy} \\ \alpha \frac{x+y}{1+\alpha xy} & 1 \end{pmatrix}$$

n-Algebraic n-Valued Groups: Examples

Continuing with the example $P(z; x, y) = -x - y + z + \alpha xyz$.

For $\alpha > 0$, the exponent of the resulting formal group is

$$\exp_P(x) = \frac{\tanh(\alpha x)}{\sqrt{\alpha}}.$$

The corresponding Q-series is

$$Q_P(x) = \frac{\sqrt{\alpha}\, x}{\tanh(\sqrt{\alpha}\, x)}.$$

So, for $\alpha = 1$, the Q-series $Q_P(x)$ gives the *L-genus* or the *signature* of a stably complex $2n$-manifold.

Symmetric n-algebraic n-valued Groups

An n-algebraic n-valued group is called *symmetric* if the polynomial

$$P(z; (-1)^n x, (-1)^n y)$$

is symmetric in the variables x, y, and z.

Let $P(z; x, y)$ define the multiplication in a symmetric n-algebraic n-valued group. Then the following hold:

- $F_0(0, y) \equiv 1$
- $F_n(x, y) = (x - y)^n$
- $\mathrm{inv}(x) = (-1)^n x$

Symmetric n-algebraic n-valued Groups

$\mathbb{G}_n$ is $\mathbb{G}(p_n)$ corresponding to the integer polynomial

$$p_n(z;(-1)^n x,(-1)^n y)=\prod_{r=1}^{n}(z-(\sqrt[n]{x}+\varepsilon^r\sqrt[n]{y})^n),$$

where ε is a primitive nth root of unity.

Symmetric 2-algebraic 2-valued Groups

For $n = 2$, each polynomial P_2 can be written in the form

$$P_2(z; -x, -y) = \sigma_1^2 - 4\sigma_2 + k_2\sigma_3 + k_8\sigma_3^2 + k_4\sigma_1\sigma_3 + k_6\sigma_2\sigma_3$$

for indeterminate graded complex coefficients k_j with $\deg k_j = -j < 0$.

Theorem (Buchstaber, Veselov, [BV19]; Kornev, 2025)

The universal symmetric 2-algebraic 2-valued group $\mathbb{G}(P_2)$ over $\mathbb{C}$ is defined over the coefficient ring

$$\mathbb{C}[k_2, k_4, k_6, k_8]/(4k_8 - k_4^2 + k_6k_2) \cong \mathbb{C}[k_2, k_4, k_6],$$

i.e., the universal family of polynomials P_2 has the form:

$$P_2(z; x, y) = \sigma_1^2 - 4\sigma_2 + 2k_2\sigma_3 + (k_4^2 - k_6k_2)\sigma_3^2 + 2k_4\sigma_1\sigma_3 + 2k_6\sigma_2\sigma_3.$$

Symmetric 3-algebraic 3-valued Groups

The general form of the polynomials P_3 is as follows:

$$\begin{aligned}\sigma_1^3 + k_0\sigma_3 + k_6\sigma_3^2 + k_{12}\sigma_3^3 + k_2\sigma_1\sigma_3 + k_4\sigma_2\sigma_3 + \\ + \ell_6\sigma_1\sigma_2\sigma_3 + \ell_4\sigma_1^2\sigma_3 + k_8\sigma_2^2\sigma_3 + \ell_8\sigma_3^2\sigma_1 + k_{10}\sigma_3^2\sigma_2,\end{aligned}$$

where the undetermined coefficients are graded by $\deg(k_i) = -i$, $\deg(\ell_j) = -j$, and each of the variables x, y, and z has degree 2.

Symmetric 3-algebraic 3-valued Groups

Theorem (Buchstaber, Kornev, 2025)

There are two classes of symmetric 3-algebraic 3-valued groups:

1. $P_3(z;-x,-y) = P_3^{\mathrm{h}}(z;-x,-y) = \sigma_1^3 - 27\sigma_3$. *We obtain the 3-valued group* $\mathbb{G}_3$.

2. $P_3(z;-x,-y) = (\sigma_1 + \ell_4\sigma_3)^3$. *The corresponding 3-valued group is obtained by the diagonal construction*

$$G \times G \xrightarrow{\mu} G \xrightarrow{\mathrm{Sym}^3 \circ \mathrm{diag}} \mathrm{Sym}^3(G)$$

from the 1-algebraic 1-valued group G, *constructed by the polynomial* $P_1(z;-x,-y) = \sigma_1 + \ell_4\sigma_3$.

Thank you!

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